Does Visual Grounding Improve Word Understanding in Language Models?

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Abstract

With the increasing adoption of Vision-Language Models (VLMs), evaluating their performance compared to language-only models has become a critical area of research. While the addition of visual modality intuitively suggests improved word understanding, recent findings by Chengxu Zhuang (2024) challenge this assumption; reporting minimal performance differences among VLMs and Language-only Models (LMs). However, their benchmarks lacked strong visual grounding, potentially biasing their results. To address this, we replicated their experiment using visually grounded benchmarks. The performance of VLMs evaluated in our experiment indicates they often outperformed GPT2, a language-only model, like Flamingo having an average accuracy of 34.5% compared to GPT2 with 27.1%. This suggests that the inclusion of visual modality can enhance a model's understanding of visually grounded language tasks, contradicting some of the conclusions by Chengxu Zhuang (2024).

1 Introduction

Large language models (LLMs) have achieved remarkable success in natural language processing (NLP), demonstrating human-like performance across a range of language tasks. However, these models are trained exclusively on text data, which fundamentally differs from how humans acquire language. Human language learning is inherently grounded in multi-modal experiences, combining inputs from vision, touch, and other sensory modalities. Vision, in particular, is hypothesized to play a crucial role in grounding language. Consequently, there has been increasing interest in VLMs. The addition of a visual modality intuitively suggests improved word understanding. Recent studies, however, provide conflicting results. For instance, Chengxu Zhuang (2024) compared VLMs and LMs in tasks of word learning and reported that the inclusion of visual modalities brought little to no benefit in some benchmarks. This finding is counterintuitive, as it contradicts the hypothesis that multimodal grounding enhances semantic understanding. A closer examination of their benchmarks reveals that these were primarily designed to evaluate language-only capabilities, potentially favoring LMs trained on large text corpora over VLMs.

To address this gap, our study replicates the work of Chengxu Zhuang (2024) by evaluating VLMs and LMs using benchmarks specifically designed to assess visual grounding. Drawing from literature (Lei Li1, 2023; Liu et al., 2022), we hypothesize that more visually grounded benchmarks will highlight performance differences between VLMs and LMs. If VLMs regularly outperform LMs on these benchmarks, it would validate the importance of visual grounding in language understanding and challenge the findings of Chengxu Zhuang (2024). By re-evaluating the role of visual modality in word learning tasks, this study aims to contribute to the growing discourse on the cognitive plausibility and utility of multi-modal models in NLP.

2 Research Question

By utilizing more visually grounded benchmarks, we aim to replicate the original experiment by Chengxu Zhuang (2024) in order to see if VLMs perform better on more visually grounded tasks compared to language-only models. If they perform better on these benchmarks, it can suggest that they possess a better understanding of word meanings. The results of this experiment could potentially challenge some of the conclusions presented by Chengxu Zhuang (2024), the initial curiosity that led us to conduct this research.

3 Related Work

This study builds upon existing research investigating the role of visual grounding in language understanding. Chengxu Zhuang (2024) explored

the impact of visual supervision on word learning across varying training dataset sizes, visual encoders, and model architectures. Their findings suggested minimal advantages for VLMs compared to Language-only Models, particularly outside of low-data regimes. This counterintuitive result motivated our investigation into the influence of benchmark selection.

In contrast, Lei Li1 (2023) compared a set of established VLMs and LMs using their newly introduced VEC (Visual and Embodied Concepts) benchmark. This benchmark was specifically designed to assess models' abilities to reason about objects' visual and embodied features. Their results demonstrated superior performance for VLMs over LMs, highlighting the importance of visual grounding for concept understanding.

Liu et al. (2022) introduced the "Spatial Commonsense" benchmark, designed to evaluate models' understanding of object scales and positional relationships. Their experiments, involving VLMs, LMs, and Image Synthesis Models (ISMs), revealed that visually grounded models, particularly ISMs, achieved the best performance in recognizing object scales. These contrasting findings underscore the need for carefully designed, visually grounded benchmarks to accurately evaluate the capabilities of VLMs, which is the focus of our work.

4 Experiment Design and Evaluation Metrics

Chengxu Zhuang (2024) reported computational resources of around 1600 hours on A100 GPUs to conduct their complete experiment across all training dataset sizes. As we had resource limitations and could not get access to trained weights of the original experiment, we conducted a limited version of the experiment in a way that results would be comparable to the original one.

In our limited experiment, we use only open-source models already trained on large datasets. One of the results by Chengxu Zhuang (2024) is that addition of visual modality brought minor to no benefit in word understanding of models, for the models that were trained on the largest training dataset size. We are benchmarking models trained only on large datasets. Observing a performance gap between language-only models and VLMs in our experiment casts doubts on this claim that visual modality brings little to no benefit in word understanding of models.

We used models and benchmarks detailed below to make our experiment both feasible and similar to the original one.

4.1 Models

The experiment includes two groups of models: a language-only model in one group and two VLMs in the other:

For the language-only model, **GPT2 medium** is used, with weights from hugging-face. This is a model with 355M parameter released by OpenAI. The original experiment included Flamingo as one of the VLMs, which is a closed-source model. After assessing the code base of the original experiment, we decided to experiment with **OpenFlamingo** to make our experiment similar to theirs. The other VLM utilized in our experiment is **CLIP**. This model includes an image encoder and a text encoder, trained to maximize the similarity of matching (image, text) pairs via a contrastive loss. We only use the text encoder module of this model which is a masked self-attention Transformer.

4.2 Benchmarks

We utilize two available benchmarks that focus on evaluation of visual understanding of language models. Both of the benchmarks were used in similar experiments to evaluate the effect of addition of visual modality. In most of the benchmarks, two subjects are compared in terms of a substantially visual quality, like shape. The first one is Spatial Commonsense which was used by Liu et al. (2022). This benchmark consists of size, height and relative position sections. Each section includes a set of yes\no questions like "Is an ant taller than a bird?" or "A boy is driving the car. Is the boy beside the car?". The other benchmark we used is ViComTe (Visual Commonsense Test) created by (Zhang et al., 2022). They used two different sources to construct the dataset: Visual Genome (Krishna et al., 2016) and Wikipedia. Their benchmark includes sections for color, shape, material, size, and co-occurrence. Six sections of this benchmark were used in our experiment; those are color, shape, material, wiki-color, wiki-shape, and wiki-material. This amounts to nine sections used in our benchmarking.

4.3 Evaluation Methods

To evaluate the models we used different methods that suited the benchmark and the model, those

are: Zero-Shot and Few-Shot prompting, adjective projection, and regression model. The details of each are explained in Appendix C.

Spatial Commonsense: To evaluate Flamingo and GPT2, we used Zero-Shot Prompting, where the model is directly asked without any additional examples to steer it. For evaluation of CLIP, adjective projection technique is used.

VL-Commonsense: These set of benchmarking datasets included a training dataset and a test dataset. For evaluation of generative models, Few-Shot prompting is used, in which we select a *subset* of training set to be included as correct examples in the prompt. The reason for including a *subset* of training set in the prompt is context length limit of models being evaluated. For evaluation of CLIP, a logistic regression model is trained. We first trained the regression models on reduced subsets of training sets to have a fair comparison with Few-Shot prompting of generative models. The trained regression models often performed very poorly so we also trained them on the complete training sets.

4.4 Evaluation metrics

For our results to be comparable to the original experiment, we evaluate performance of models using top-1 accuracy. In case of evaluating the output of generative models, our strategy is dependent on the benchmark used. If the prompt is asking a yes\no question, then we check for existence of yes\no in the output. Sometimes the model outputs an ambiguous sentence without either yes or no but semantically correct. We manually analyzed those outputs. The details are in Appendix D. In other cases where the generative model is asked to output the quality type of a subject, for example to complete "The color of snow is", we check the exact match of the first generated word with the first label in set of correct labels, hence a top-1 accuracy. Top-1 accuracy was also used to measure the accuracy of trained regression model.

5 Results

The results of the experiment are summarized in Figure 1 in Appendix B. In 8 out of 9 different benchmarkings, there was at least one VLM that performed better than the language-only model, i.e. GPT2. The average accuracy of different models across of benchmarkings are as follows: CLIP: 40.9%, CLIP (*): 6.8%, Flamingo: 34.5%, GPT2: 27.1%. This suggests that some VLMs, whose training set includes visual modality, might have better understanding of visual aspects of the world, since these benchmarks are trying to gauge visual understanding of models. When the regression model on top of CLIP is trained on a reduced subset of the training set, it learned to predict a single label for all test samples, which indicates the insufficiency of training on reduced training sets. When the complete training set were used, it produced higher test results, except in the case of VL-commonsense::wiki-color.

6 Conclusion

In this study, we sought to replicate the experiment of Chengxu Zhuang (2024) by focusing on the evaluation of models using visually grounded benchmarks. While the original study reported minimal benefits of visual modality in word understanding, their findings were potentially limited by benchmarks lacking strong visual grounding. Our experiment, though resource-constrained, demonstrated that VLMs like Flamingo often outperformed GPT2. These results suggest that the inclusion of visual modalities enhances a model's understanding of words.

Our approach and results can be replicated using the provided code base. We recommend the development of improved multi-modal benchmarks to further assess and refine the capabilities of VLMs in capturing nuanced visual and linguistic interactions.

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Appendix

A: Description of Ineffectiveness of Benchmarks in Original Experiment

We believe that benchmark from (Chengxu Zhuang, 2024) paper cannot correctly evaluate word meaning for an experiment focused on the addition of visual modality. Here we describe it in detail to show it considers word meaning only from a linguistic and statistical point of view:

Word similarity: human judgments on how much two **words** are related.
Brain-Response prediction: predict the brain response to one **sentence** using its hidden representations.
Lexical relation prediction: lexical-relation categories of two **words**: synonymy, antonym, hypernymy, part-whole meronymy, and random.
Semantic feature prediction : ask human annotators to write down features of the **word**.

B: Accuracy Chart of Benchmarking

Accuracies of different models are summarized in the Figure 1.

C: Description of Evaluation Methods Used

Here is a description of different evaluation methods used in this experiment:

Adjective Projection is the method used by Zhang et al. (2022) to evaluate CLIP model. In this method, to compare subjects (e.g., "ant" and "bird") along an adjective scale (e.g., "tallness"), word embeddings are compare. First, the adjective scale vector is calculated by $\overrightarrow{tall} - \overrightarrow{short}$ ($\overrightarrow{tall}$ is the embedding of "tall"). Then, the projection score on the "tallness" scale for a subject, like $\overrightarrow{ant}$, is calculated by $cos(\overrightarrow{adj_scale}, \overrightarrow{ant})$. In the last step, we compare projection scores of different subjects.

$$a = cosine_sim(\overrightarrow{adj_scale}, \overrightarrow{ant})$$
$$b = cosine_sim(\overrightarrow{adj_scale}, \overrightarrow{bird})$$

if $a > b$, then ant is taller than bird.

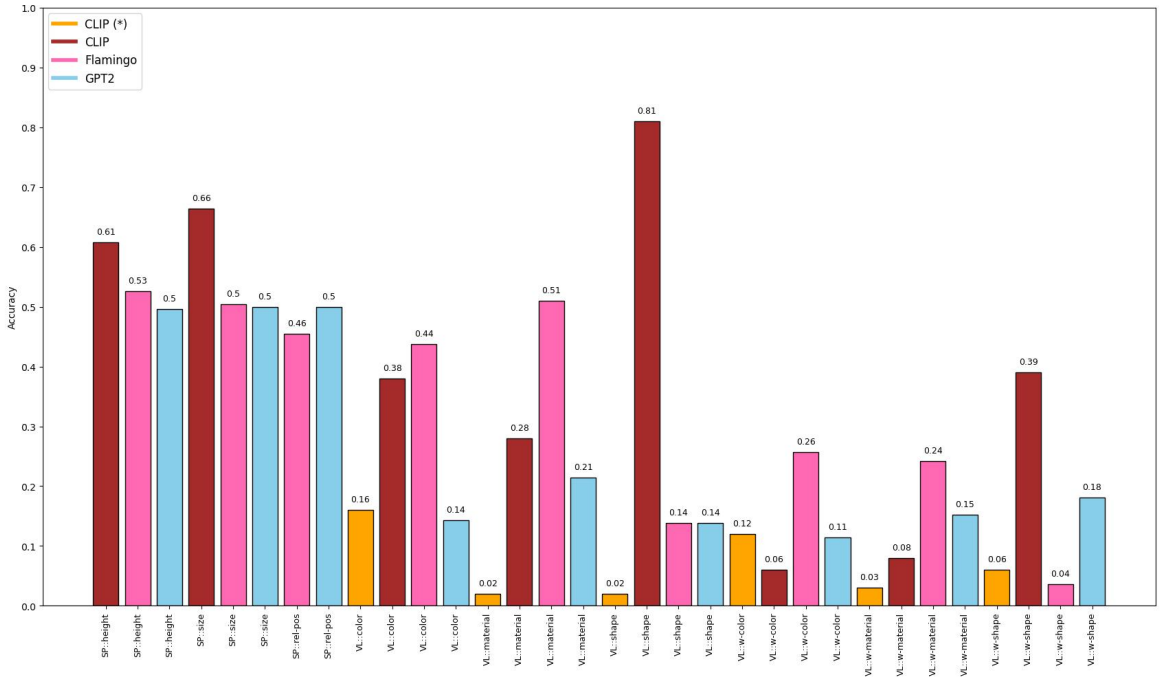


Figure 1: Accuracies of different models benchmarkings.
The abbreviations in the chart are interpreted as follows: **SP::x** refers to the *x section* of Spatial Commonsense; **VL::y** denotes the *y section* of VL-Commonsense; and **VL::w-color** corresponds to the *wiki-color section* of VL-Commonsense. The y-axis represents model performance, and the x-axis lists the various benchmarks. **CLIP(*)** is regression model trained on reduced training set, while **CLIP** is regression model trained on complete set.

Zero-Shot Prompting is a technique where the model is asked the question without being shown any specific examples beforehand.

> Prompt: "Answer this question with yes or no: Is an ant taller than a bird?"

Few-Shot Prompting is a technique that provides the model with a few correct examples, in the early query phase. By doing so the model manages to learn immediately relevant patterns.

For example, the prompt for color benchmark is produced by this routine. From the training set of color section, three colors for each color type are randomly chosen, then the selected colors are turned into sentences. Finally, the produced sentences are shuffled.

> The color of patio is gray. The color of sunlight is yellow. The color of brake light is red. The color of flush handle is silver. The color of {subject} is

Regression model on top of CLIP: For evaluation of CLIP, a logistic regression model is trained. The input to this model are embeddings out of text encoder module of CLIP. We fed samples from training set to text encoder of CLIP and extracted the pooled embeddings for each sample, and then used these along with the classes of each sample in training set to train the regression model. The trained model often performed poorly, partly because it was trained on a limited training set consisting of only 21 to 48 samples. So we also trained the regression model on the complete training sets from VL-commonsense which included at least 222 samples across any section of VL-commonsense.

D: Samples of Ambiguous Responses Generated in Benchmarking

Here are some examples of ambiguous answers produced by generative models. Our simple logic can not distinguish if the response must be counted as correct, so we log them to be analyzed manually later. The simple logic is to check the existence of yes\no in the generated response.

Question:"A boy drives the car. Is the boy inside the car?" **reply**: "the boy is inside the car."
Question: "A man is watching the firework. Is the man beside the firework?" **reply**: "the man is watching the firework. answer: the man is watching the firework. answer"